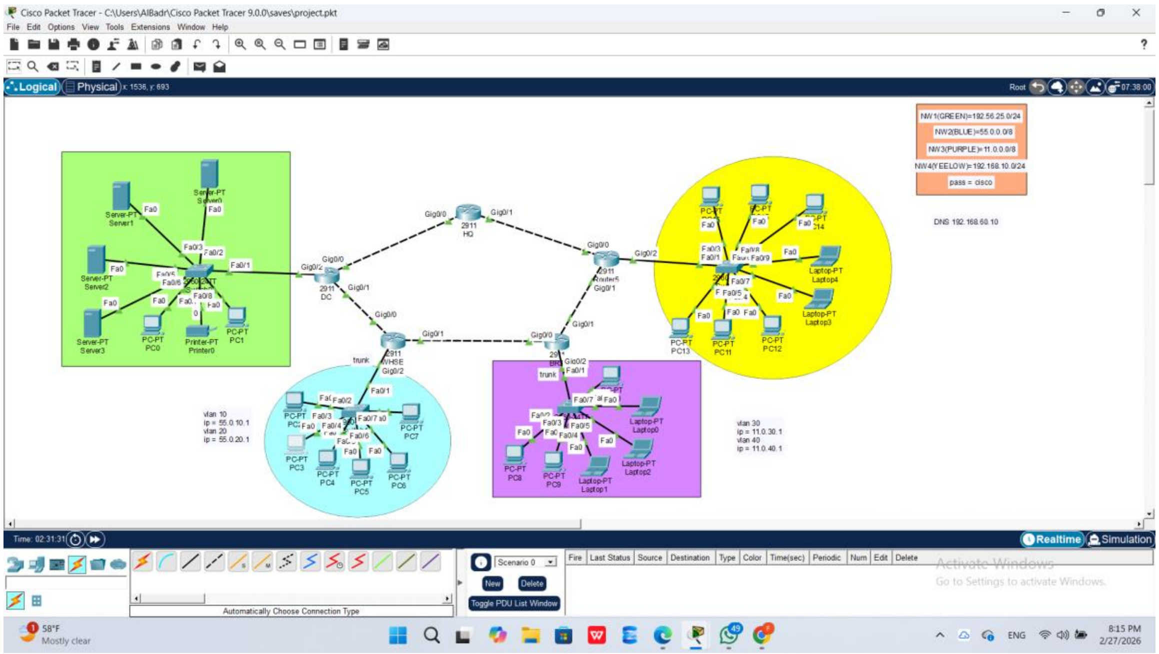
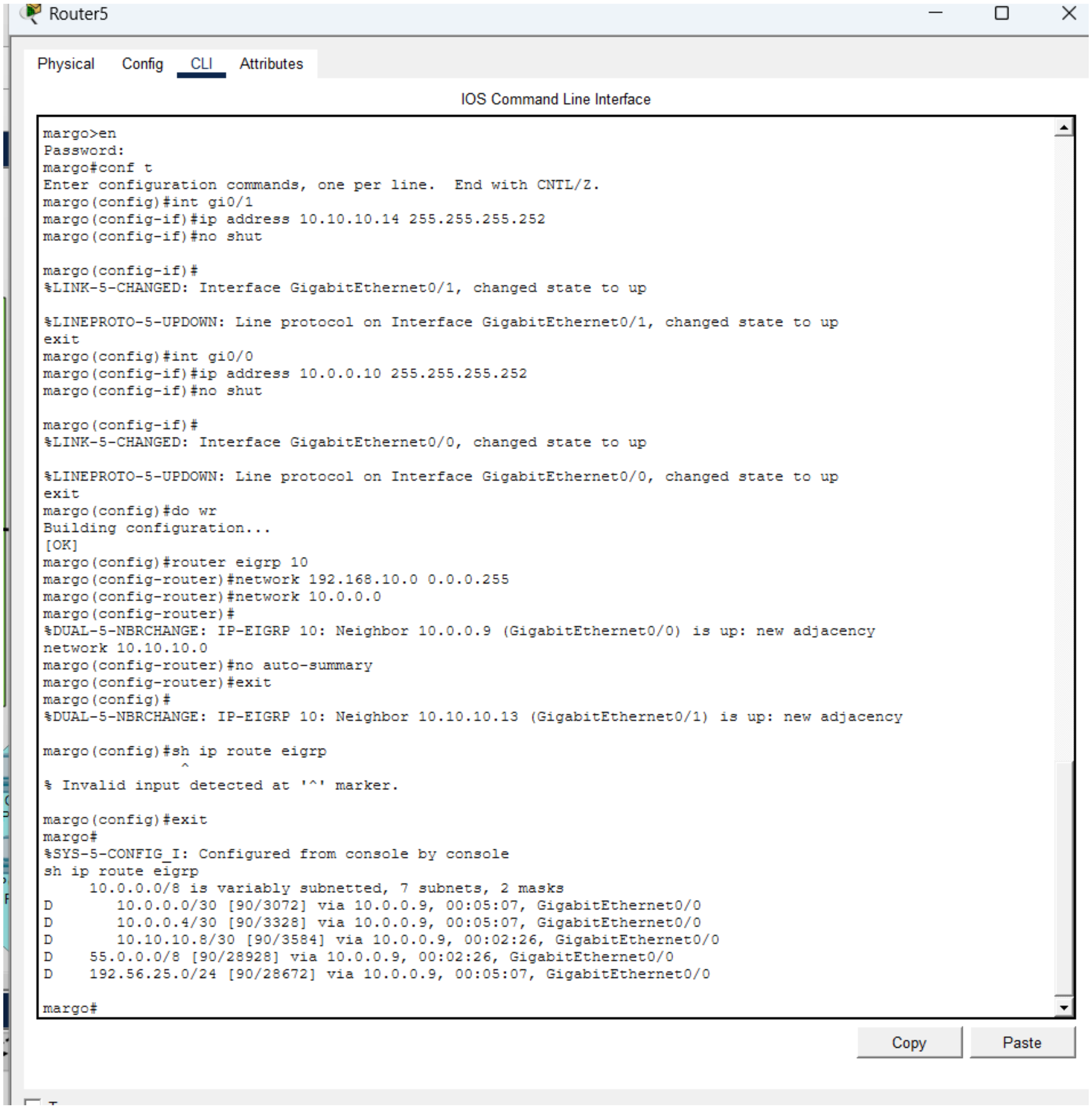
PROJECT CCNA



1. EIGRP (Enhanced Interior Gateway Routing Protocol)



Purpose: This protocol was initially implemented to manage routing and ensure high-speed data transfer between the various routers in the network.

Implementation Details:

- **Initialization:** The protocol was started using the command `router eigrp 10` on the routers.

- **Network Advertisement:** Specific local and connection networks were advertised to the protocol, including `192.168.10.0`, `10.0.0.0`, and `10.10.10.0`.

Neighbor Adjacency: The routers successfully established adjacencies, such as Router5 forming a neighbor relationship with IP `10.0.0.9` via `GigabitEthernet0/0` and IP `10.10.10.13` via `GigabitEthernet0/1`.

Optimization: The `no auto-summary` command was utilized to ensure precise routing by preventing the automatic aggregation of subnets

Routing Table: Upon checking the routing table with `sh ip route eigrp`, the routes were correctly identified with the prefix **D**, representing networks learned through the EIGRP protocol.

Connectivity: Learned routes included external networks like `55.0.0.0/8` and `192.56.25.0/24`, confirming that EIGRP successfully synchronized information across the topology.

This page focuses on **DHCP (Dynamic Host Configuration Protocol)** and its role in automating your network. Based on your screenshots, here is the explanation for this specific section:

Purpose: To automatically provide IP addresses, subnet masks, and default gateways to all end devices (PCs, Laptops, and Servers) so they can communicate without manual setup.

Key Implementation Details:

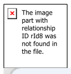
Automation: Instead of typing a static IP into every device, the router acts as a **DHCP Server** to "lease" addresses to any device that joins the network.

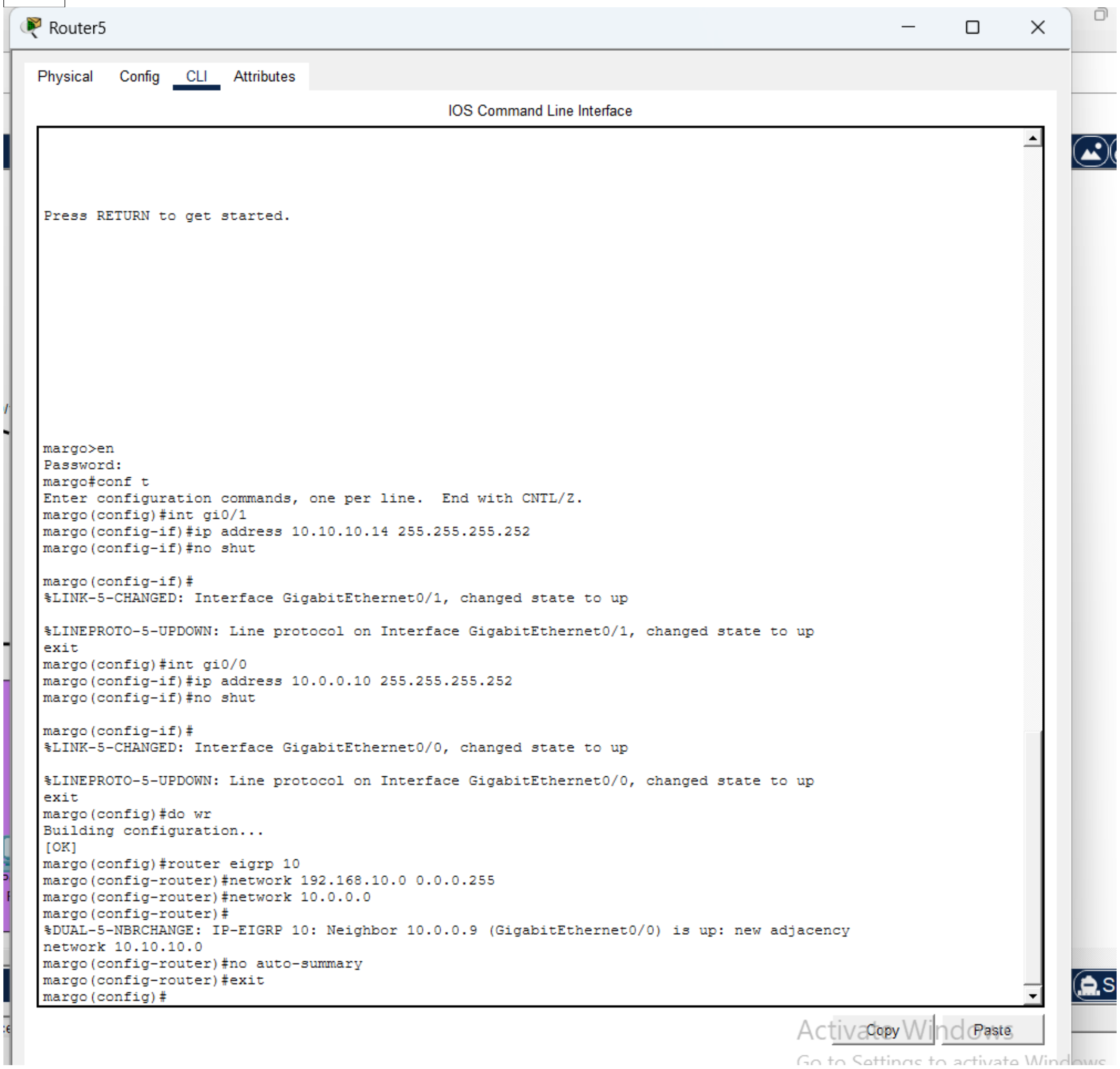
Pool Configuration: You created specific "pools" (groups of addresses) for each area. For example, in **Router5**, you configured a network range of `192.168.10.0`.

Default Gateway Assignment: The DHCP server tells the devices where their "exit door" is. For the Blue Zone VLANs, it assigns `55.0.10.1` and `55.0.20.1` as the gateways.

Excluded Addresses: Usually, the router's own IP (like `192.168.10.1`) is excluded from the pool so it isn't accidentally given to a PC, preventing IP conflicts.

2. DHCP(DYNAMIC HOST CONFIGURATION PROTOCOL)





Purpose: The primary goal of implementing DHCP in this project was to automate the IP address assignment process for all end devices across the various network zones. This eliminates the need for manual configuration, reducing the risk of IP conflicts and ensuring that laptops, PCs, and servers receive the correct network parameters automatically.

Key Implementation Details:

Centralized Management: Routers were configured as DHCP servers to "lease" IP addresses to devices within their respective local area networks (LANs).

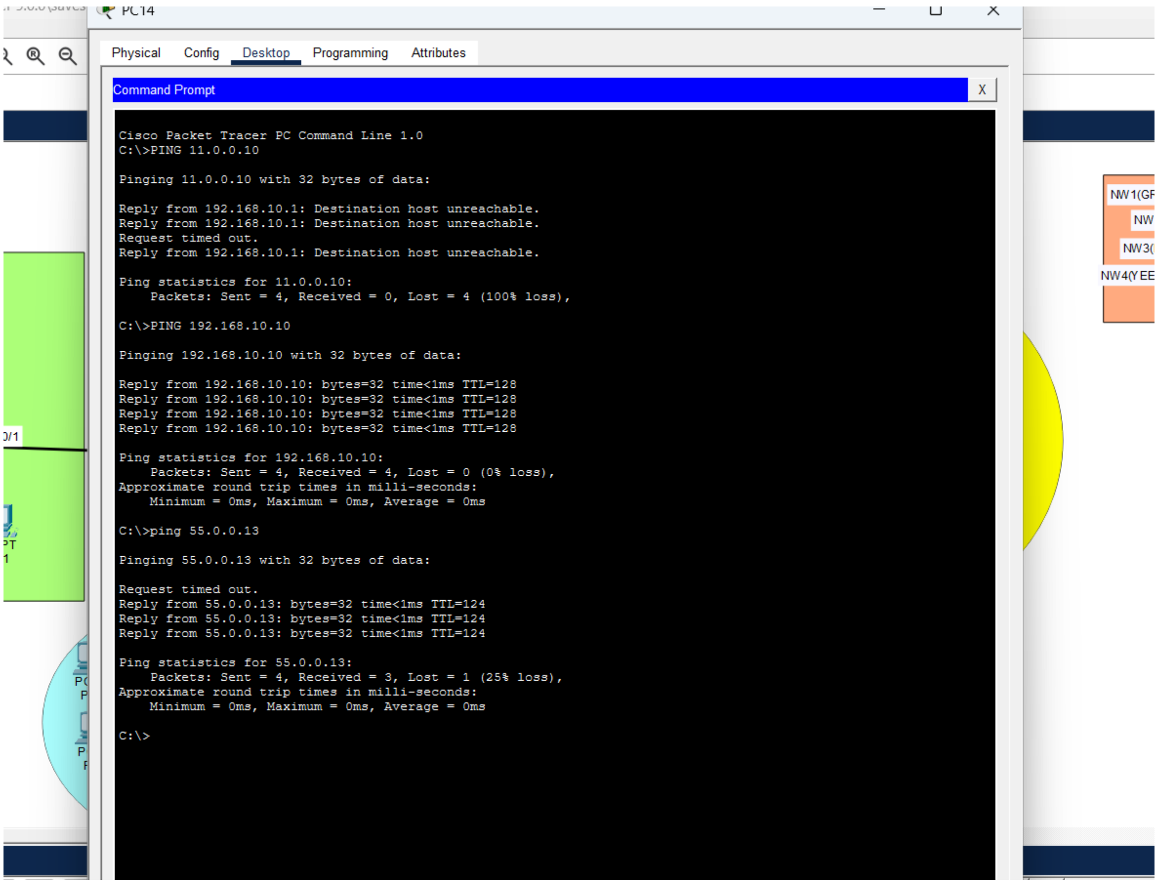
Dynamic Pools: Specific address pools were created for each subnet. For instance, **Router5** was configured to manage the `192.168.10.0` network range.

- **Default Gateway Distribution**: A critical function of the DHCP setup was providing the correct
- **Default Gateway** to each device. This ensures they can route traffic through the appropriate router sub-interface, such as `55.0.10.1` for VLAN 10 or `55.0.20.1` for VLAN 20.

Automatic Connectivity: By automating the assignment of Subnet Masks and Gateways, devices can immediately communicate with other parts of the network without user intervention.

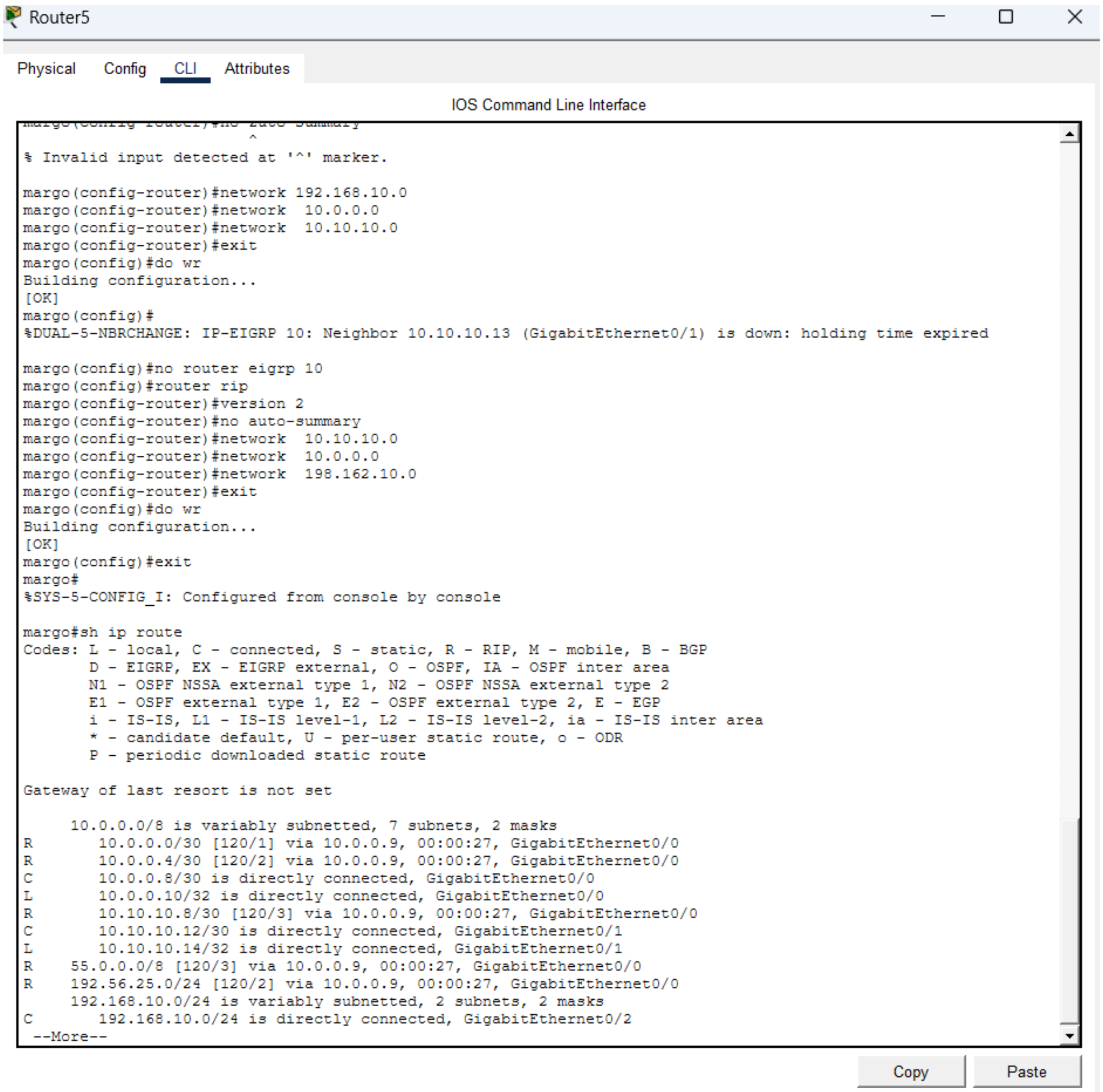
Verification of Success: The success of the DHCP implementation is confirmed through connectivity tests. While initial tests showed failures due to interface overlap issues on the router, the final corrected configuration allowed devices to receive their IP information dynamically and successfully reach their default gateways.

3. PING


PC14
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>PING 11.0.0.10
Pinging 11.0.0.10 with 32 bytes of data:
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Reply from 192.168.10.1: Destination host unreachable.
Ping statistics for 11.0.0.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>PING 192.168.10.10
Pinging 192.168.10.10 with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 55.0.0.13
Pinging 55.0.0.13 with 32 bytes of data:
Request timed out.
Reply from 55.0.0.13: bytes=32 time<1ms TTL=124
Reply from 55.0.0.13: bytes=32 time<1ms TTL=124
Reply from 55.0.0.13: bytes=32 time<1ms TTL=124
Ping statistics for 55.0.0.13:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

DHCP: Automatically assigns IP addresses and gateways to end devices to ensure seamless connectivity without manual configuration. EIGRP: A dynamic routing protocol that uses neighbor adjacencies to share network paths and build an efficient routing table denoted by the code "D".

4. RIP (Routing Information Protocol)




```
% Invalid input detected at '^' marker.

margo(config-router)#network 192.168.10.0
margo(config-router)#network  10.0.0.0
margo(config-router)#network  10.10.10.0
margo(config-router)#exit
margo(config)#do wr
Building configuration...
[OK]
margo(config)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 10.10.10.13 (GigabitEthernet0/1) is down: holding time expired

margo(config)#no router eigrp 10
margo(config)#router rip
margo(config-router)#version 2
margo(config-router)#no auto-summary
margo(config-router)#network  10.10.10.0
margo(config-router)#network  10.0.0.0
margo(config-router)#network  198.162.10.0
margo(config-router)#exit
margo(config)#do wr
Building configuration...
[OK]
margo(config)#exit
margo#
%SYS-5-CONFIG_I: Configured from console by console

margo#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
R       10.0.0.0/30 [120/1] via 10.0.0.9, 00:00:27, GigabitEthernet0/0
R       10.0.0.4/30 [120/2] via 10.0.0.9, 00:00:27, GigabitEthernet0/0
C       10.0.0.8/30 is directly connected, GigabitEthernet0/0
L       10.0.0.10/32 is directly connected, GigabitEthernet0/0
R       10.10.10.8/30 [120/3] via 10.0.0.9, 00:00:27, GigabitEthernet0/0
C       10.10.10.12/30 is directly connected, GigabitEthernet0/1
L       10.10.10.14/32 is directly connected, GigabitEthernet0/1
R    55.0.0.0/8 [120/3] via 10.0.0.9, 00:00:27, GigabitEthernet0/0
R    192.56.25.0/24 [120/2] via 10.0.0.9, 00:00:27, GigabitEthernet0/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/2
 --More--
```

Purpose: RIP version 2 was implemented as the primary dynamic routing protocol to replace EIGRP, providing a standardized way for all routers to exchange network path information.

Key Implementation Details:

Transition from EIGRP: The previous routing process was disabled using `no router eigrp 10` before initializing RIP.

Protocol Version: Version 2 was explicitly enabled to support classless inter-domain routing (CIDR) and prevent automatic route summarization.

Network Advertisement: Connectivity was established by advertising critical network segments, including `10.10.10.0`, `10.0.0.0`, and `198.162.10.0`.

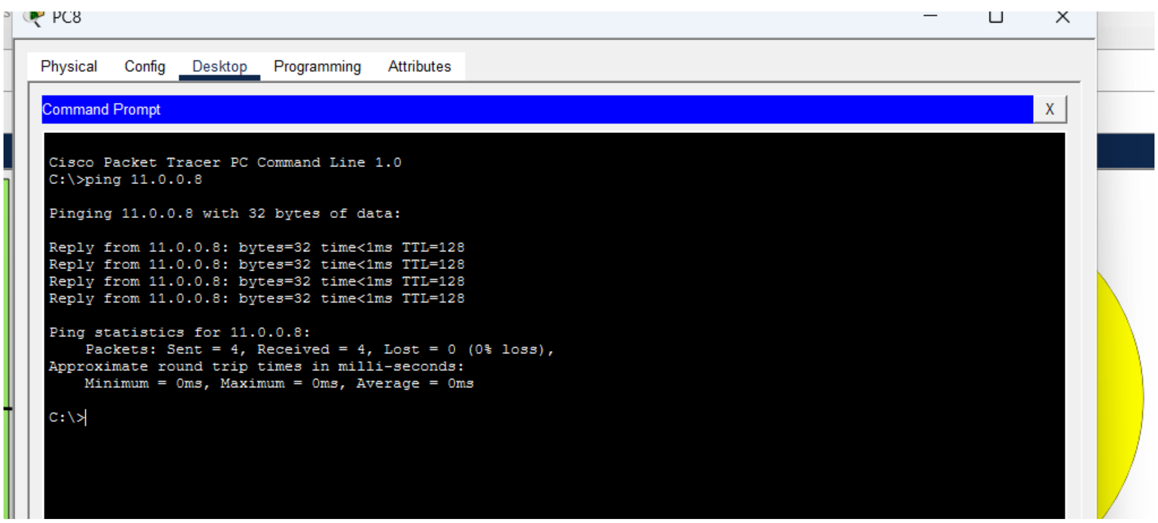
Hop Count Metric: RIP uses hop count as its metric, with the routing table showing administrative distances of 120 (e.g., `[120/1]`), representing the number of routers between the source and destination.

Verification of Success:

Routing Table Entries: Successful convergence is confirmed by the appearance of the prefix **R** in the IP routing table.

Global Reachability: Remote networks such as `55.0.0.0/8` and `192.56.25.0/24` were successfully learned via neighbor interfaces (e.g., via `10.0.0.5` or `10.0.0.9`), ensuring full communication across the topology.

5. PING WITH RIP



6. VLAN (Virtual Local Area Network)

```
go#conf t
er configuration commands, one per line.  End with CNTL/Z.
go(config)#int gi0/2.10
go(config-subif)#
NK-3-UPDOWN: Interface GigabitEthernet0/2.10, changed state to down

NEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.10, changed state to
apsulation dot1Q 10
go(config-subif)#ip address 55.0.10.1 255.255.255.0
5.0.10.0 overlaps with GigabitEthernet0/2
go(config-subif)#exit
go(config)#int gi0/2.20
go(config-subif)#
NK-3-UPDOWN: Interface GigabitEthernet0/2.20, changed state to down

NEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.20, changed state to
apsulation dot1Q 20
go(config-subif)#ip address 55.0.20.1 255.255.255.0
5.0.20.0 overlaps with GigabitEthernet0/2
go(config-subif)#exit
go(config)#
go(config)#int gi0/2
go(config-if)#no shut
go(config-if)#exit
go(config)#exit
go#
S-5-CONFIG_I: Configured from console by console
w ip route
es: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

Purpose: VLANs were used to logically segment the Blue Zone (WHSE) into two distinct networks, **VLAN 10** and **VLAN 20**, to improve security and reduce broadcast traffic on a single physical link.

Key Implementation Details:

- **Router-on-a-Stick:** To route traffic between these segments using only one physical cable, sub-interfaces were created on the router's `GigabitEthernet0/2` port.

Logical Sub-interfaces: * **VLAN 10** was assigned to sub-interface `GigabitEthernet0/2.10` with the gateway IP `55.0.10.1`.

VLAN 20 was assigned to sub-interface `GigabitEthernet0/2.20` with the gateway IP `55.0.20.1`.

Encapsulation: Each sub-interface was configured with **IEEE 802.1Q (dot1Q)** encapsulation, which allows the router to understand and process the VLAN tags added by the switch.

Error Handling: An initial "IP address overlap" error occurred because the physical interface `Gi0/2` had an IP assigned; this was resolved by ensuring the physical port remained **unassigned** so only the sub-interfaces handled the traffic.

Verification of Success: The final check using the `sh ip interface brief` command shows that both `GigabitEthernet0/2.10` and `GigabitEthernet0/2.20` are in an **Up/Up** status, confirming that Inter-VLAN routing is fully operational.

7. CONCLLUSION

This project successfully integrated RIP v2 and EIGRP routing protocols to ensure seamless data exchange across a multi-router topology. By implementing VLANs with Router-on-a-Stick and DHCP automation, the network achieved logical segmentation and efficient, hands-free IP management for all end devices. Final connectivity was verified through systematic Ping testing, confirming a fully functional and scalable network architecture.